# CODE FILE

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-- STUDENT COURSE REGISTRATION SYSTEM
-- DBMS Mini Project | MySQL 8.0+

CREATE DATABASE IF NOT EXISTS student_course_db;
USE student_course_db;

-- Drop dependent objects first if rerunning the script
DROP VIEW IF EXISTS StudentCourseView;
DROP TABLE IF EXISTS Registrations;
DROP TABLE IF EXISTS Courses;
DROP TABLE IF EXISTS Students;

-- 1. Students table
CREATE TABLE Students (
    student_id INT PRIMARY KEY AUTO_INCREMENT,
    student_name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    phone VARCHAR(15),
    semester INT NOT NULL,
    CHECK (semester BETWEEN 1 AND 8)
);
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-- 2. Courses table
CREATE TABLE Courses (
    course_id INT PRIMARY KEY AUTO_INCREMENT,
    course_name VARCHAR(100) NOT NULL,
    credits INT NOT NULL,
    fee DECIMAL(10,2) DEFAULT 0,
    CHECK (credits > 0),
    CHECK (fee >= 0)
);

-- 3. Registrations table
CREATE TABLE Registrations (
    registration_id INT PRIMARY KEY AUTO_INCREMENT,
    student_id INT NOT NULL,
    course_id INT NOT NULL,
    registration_date DATE NOT NULL,
    status VARCHAR(20) DEFAULT 'Registered',

    CONSTRAINT fk_registration_student
        FOREIGN KEY (student_id)
        REFERENCES Students(student_id),

    CONSTRAINT fk_registration_course
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    FOREIGN KEY (course_id)
    REFERENCES Courses(course_id),

  CONSTRAINT uq_student_course
    UNIQUE (student_id, course_id),

  CHECK (status IN ('Registered', 'Completed', 'Cancelled'))
);

-- 4. Insert sample students
INSERT INTO Students
(student_name, email, phone, semester)
VALUES
('Aman Sharma', 'aman@gmail.com', '9876543210', 3),
('Priya Singh', 'priya@gmail.com', '9876543211', 3),
('Rahul Kumar', 'rahul@gmail.com', '9876543212', 5),
('Neha Verma', 'neha@gmail.com', '9876543213', 5),
('Karan Singh', 'karan@gmail.com', '9876543214', 1);

-- 5. Insert sample courses
INSERT INTO Courses (course_name, credits, fee)
VALUES
('Database Management System', 4, 5000),
('Web Development', 3, 4000),
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('Data Structures', 4, 4500),
('Computer Networks', 3, 3500);

-- 6. Insert registrations
INSERT INTO Registrations
(student_id, course_id, registration_date, status)
VALUES
(1, 1, '2026-10-01', 'Registered'),
(1, 2, '2026-10-02', 'Registered'),
(2, 1, '2026-10-03', 'Completed'),
(3, 3, '2026-10-04', 'Registered'),
(4, 4, '2026-10-05', 'Registered'),
(5, 2, '2026-10-06', 'Registered');

-- 7. Display tables
SELECT * FROM Students;
SELECT * FROM Courses;
SELECT * FROM Registrations;

-- 8. INNER JOIN
SELECT s.student_name, c.course_name,
    r.registration_date, r.status
FROM Students s
INNER JOIN Registrations r
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   ON s.student_id = r.student_id
INNER JOIN Courses c
   ON r.course_id = c.course_id;

-- 9. LEFT JOIN: include students without registrations
SELECT s.student_name, c.course_name
FROM Students s
LEFT JOIN Registrations r
   ON s.student_id = r.student_id
LEFT JOIN Courses c
   ON r.course_id = c.course_id;

-- 10. Count registrations by course
SELECT c.course_name,
      COUNT(r.registration_id) AS total_registrations
FROM Courses c
LEFT JOIN Registrations r
   ON c.course_id = r.course_id
GROUP BY c.course_id, c.course_name;

-- 11. HAVING example
SELECT c.course_name,
      COUNT(r.registration_id) AS total_students
FROM Courses c
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JOIN Registrations r
    ON c.course_id = r.course_id
GROUP BY c.course_id, c.course_name
HAVING COUNT(r.registration_id) > 1;

-- 12. Search students
SELECT *
FROM Students
WHERE student_name = 'Aman Sharma';

-- 13. Update registration status
UPDATE Registrations
SET status = 'Completed'
WHERE registration_id = 1;

-- 14. Create a view
CREATE VIEW StudentCourseView AS
SELECT s.student_id, s.student_name,
       c.course_name, c.credits, c.fee,
       r.registration_date, r.status
FROM Students s
JOIN Registrations r
    ON s.student_id = r.student_id
JOIN Courses c
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    ON r.course_id = c.course_id;

SELECT * FROM StudentCourseView;

-- 15. Create index
CREATE INDEX idx_student_name
ON Students(student_name);

-- 16. Query execution plan
EXPLAIN
SELECT *
FROM Students
WHERE student_name = 'Aman Sharma';

-- 17. Transaction demonstration
START TRANSACTION;

INSERT INTO Registrations
(student_id, course_id, registration_date, status)
VALUES
(2, 3, '2026-10-07', 'Registered');

COMMIT;
```

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-- 18. Verify final data

SELECT * FROM Registrations;

SELECT * FROM StudentCourseView;
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